Reference Event 2: The Faulty Content Update of 19 July 2024

Characterization Against the Information Technology Sector Design Basis

DBA-IT-RE-2 — Reference Event Instrument, DBA-IT Series

Lineage: Issued under the Information Technology Sector Framework v0.5; DBA-MA Parent Framework; conferred on the proposal of Software, Hardware and Product Supply Chain (DBA-IT-SW-FC3)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the faulty content update to the CrowdStrike Falcon sensor of 19 July 2024 back through DBA-IT-SW-FC3 (*Software, Hardware and Product Supply Chain*), whose category of a defective update delivered through the trusted channel it characterizes (§5.1), at the class's second reference position, the endpoint security agent (§1.3). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Information Technology Sector Framework v0.5, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 14 October 2026, on the proposal of *Software, Hardware and Product Supply Chain*, as an accident for the class's availability chain. Section references of the bare form §N are to that class at v0.1, the version this instrument read.

The scope runs from the introduction of the template type in February 2024, through the release and reversion of the content on 19 July, the recovery of the hosts to 29 July, and the vendor's root-cause analysis of 6 August, to the House hearing of 24 September 2024 and the operating-system vendor's announcement of 19 November 2024. The technical record is closed. The consequences for one airline remain in litigation, and that part of the record is open.

No individual is named; the vendor's witness and the members of the committee are named by role, and the vendor, the operating-system vendor, the airline and the public bodies as institutions. The event was an accident. This instrument makes no finding of fault, and states the litigation's claims as the parties' own.

The sensor, its content and its channel lie within this class. The answering points, hospitals, banks, airlines and federal agencies that ran it are consuming facilities of other classes and sectors. The Foundational Definitions are cited by designation, FD-REG holding their versions: FD-LD (*Layer Decomposition and Determinism*) bears on Section 6.3.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the vendor's *Preliminary Post Incident Review* of July 2024, its *Channel File 291 Incident Root Cause Analysis* of 6 August 2024 and its remediation and guidance hub; the operating-system vendor's count of 20 July 2024 and its announcement of 19 November 2024; the record of the House Committee on Homeland Security's hearing of 24

September 2024 (Serial No. 118-81), read for the members' and the witness's statements of that day; and the Congressional Research Service's *CrowdStrike IT Outage: Impacts to Public Safety Systems and Considerations for Congress* (IF12717, version of 4 December 2024). The airline's figures are as reported of the Department of Transportation's review. No non-public material is used.

2.2 Where the Sources Differ

They do not differ on the technical facts. On the consequences, the members' statements at the hearing are general ("Their call to 911? It wouldn't go through"), and the Research Service gives the places; both are stated as their authors'. The date of the stress test, 5 March 2024, is the preliminary review's; the root-cause analysis does not date it.

3. The Event

3.1 The Sensor and Its Channels

The sensor runs in the operating system's kernel. It is changed by two kinds of content. Sensor content ships with each sensor release, and customers could hold their sensors at the current release or one or two behind it. Rapid response content is delivered from the vendor's cloud as configuration, which the preliminary review says "is not code or a kernel driver", and is read by an interpreter in the sensor; the vendor's witness told the committee the company released "10 to 12 of these updates, content updates, every single day". The customer's version policy governed sensor releases and did not govern rapid response content.

3.2 The Latent Defect

With sensor 7.11 in February 2024 the vendor introduced a new template type for inter-process communication. The type defined twenty-one input fields; the sensor code that supplied inputs to it supplied twenty. The vendor's content validator assessed new content against twenty-one. The interpreter had no runtime check on the bounds of the array it read. The type passed a stress test in staging on 5 March 2024, and the test content used a wildcard in the twenty-first field, so the fault was never exercised (root-cause analysis; preliminary review).

3.3 The Release

On 19 July 2024 at 04:09 UTC the vendor released content whose template instance used a non-wildcard value in the twenty-first field. It went to all customers in one session. Windows hosts running sensor 7.11 or later that were online and received it read beyond the array and crashed, and on restart crashed again; Mac and Linux hosts were not affected. The vendor reverted the content at 05:27 UTC, seventy-eight minutes later. The operating-system vendor counted about 8.5 million Windows devices affected, less than one percent of all Windows machines.

3.4 The Recovery

A host that could reach the channel long enough after restart could receive the reverted file; many could not. The vendor's guidance was to restart, and failing that to boot into recovery and delete the faulty file by hand, which on an encrypted disk required its recovery key; cloud machines were recovered by detaching and repairing their disks or rolling back to a snapshot. The vendor reported about 99 percent of Windows sensors online by the evening of 29 July, measured against the week before.

3.5 The Consuming Facilities

The Research Service records that in Phoenix calls to 911 were answered but caller information was recorded by hand because computer-aided dispatch was affected; that in Portland calls were answered manually; that the Alaska State Troopers said 911 systems were not operating correctly and posted other numbers; and that police in Middletown, Ohio, said 911 calls were not being received. At the hearing the subcommittee's chairman said hospitals saw disruptions in their medical systems, banks experienced downtime in transaction processing, thousands of flights were grounded or cancelled worldwide, and federal agencies were temporarily unable to access certain data. Delta Air Lines cancelled about 7,000 flights over five days, by the figures reported when the Department of Transportation closed its review of the airline's customer service in 2026 without seeking penalties; the airline's claims against the vendor, of which gross negligence and computer trespass survived a motion to dismiss in May 2025, are in litigation.

3.6 What Changed

The vendor committed to a staggered deployment of rapid response content beginning with a canary, to give customers control over where and when it is deployed, and to independent third-party review of its code and processes; its root-cause analysis added a compile-time field-count check, a runtime bounds check and tests with non-wildcard values in each field. On 19 November 2024 the operating-system vendor announced a recovery feature to apply fixes from its update service to machines that cannot boot, and capabilities to let security products run outside the kernel.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-4) and effect elements (E-1 to E-5), as Table 4-1 sets out. Composed, they are the class's defective update through the trusted channel (§5.1) at the endpoint security agent (§1.3), in Chain A, immediate and unselective, with hosts the channel could not correct.

Table 4-1. Event elements and their mapping. Section references are to *Software, Hardware and Product Supply Chain* v0.1.

No.	What happened	Class provision	Section

I-1	Twenty-one fields defined, twenty supplied; validator and interpreter unguarded	Test in the configuration that runs (§5.1, §7.2)	3.2
I-2	Template type tested as a type with a wildcard in the twenty-first field	Determinism posture (§7.2)	3.2
I-3	Content released to all customers in one session	Staging (§4.4)	3.3
I-4	Customer version policy did not govern content	Consumer's authority over every channel (§4.5)	3.1
E-1	About 8.5 million hosts crash in a seventy-eight-minute window	Reach; propagation time (§3.3)	3.3
E-2	Hosts crash before the channel can correct them; recovery by hand, with recovery keys	Recovery independent of the channel (§4.6)	3.4
E-3	About 99 percent of sensors online in ten days	Time to recovery (§3.3)	3.4
E-4	911 call handling manual or interrupted; hospitals, banks, flights, agencies disrupted	Consequence in other sectors (§2.4, §8.1)	3.5
E-5	One airline's operation takes five days to recover	Time to recovery (§3.3, §4.6)	3.5

5. Tier Determination

5.1 The Four Characteristics

The event was not an attack. Intentionality, persistence, precision and denial are not present.

5.2 Determination

The event is non-adversarial. It realizes the class's §5.1 in Chain A across the release's reach, and is the class's reference case for that chain. It does not realize §6.3, whose activation is deliberate, but it supplies the figures for that scenario's consequence half. That is the tier determination returned for the DBA Reference Event Register. Emergency Services, Healthcare and Public Health, Financial Services, Transportation and Government Facilities are proposed as impacted sectors, on the record of Section 3.5.

6. Characterization Against the Design Basis

6.1 §5.1, §3.2 and §4.6 The Class's Account of the Event

Held, with a gap. The account holds in every figure checked: the release and reversion times, the 8.5 million devices, the three faults together, the stress test, the wildcard, ten to twelve content updates a day, and about 99 percent online by 29 July. Two points are qualified and go to the class as Erratum 2 of Project 11's note of 14 October 2026 (Information Technology batch): hosts that received the content did not all fail beyond the channel's reach, since some received the reverted file on restart, and §3.2 and §4.6 should say "many" rather than imply all; and the 911 consequence should be stated from the Research Service's places rather than from the

members' general statements, with the hearing's figures attributed to the members who gave them.

6.2 §4.4 Staging and the First Ring

Held, with a gap. The content went to every customer in one session, and the class's requirement that every behavior-changing channel be staged is the vendor's own commitment since. The class declares the first ring's loss as a residual (§6.8) and asks the vendor to state its first ring. It does not say who may be in it, and on 19 July answering points were among the hosts the one session reached. Finding F-1.

6.3 §5.1 and §7.2 The Configuration That Was Tested

Held. The template type was tested; the configuration that ran on 19 July was not. The class's reading, and the author's determination of 8 October 2026 for FD-LD (*Layer Decomposition and Determinism*) §4.3 at its next revision, were written from this event and read it.

6.4 §4.5 The Consumer's Authority Over What Enters

Held. A customer could hold the sensor behind the current release and could not hold the content, which was the channel that failed. The class names exactly this, and the vendor has since given customers the control.

6.5 §4.6 Recovery That Does Not Depend on the Channel

Held, with a gap. Recovery was by hand, host by host, with recovery keys, and the class's requirement of an independent path is the event's lesson. The event adds that the vendor's ten days measure sensors online, while the consuming enterprise's recovery is of its service: an airline whose hosts returned took five days to restore its operation. Finding F-2.

6.6 §3.3 Privilege as a Parameter

Held, with a gap. The class states privilege as a parameter that sets whether a host can be recovered without hands at the console. The response of 19 November 2024 shows privilege is also a design choice that can be lowered, by moving the product out of the kernel, and that a platform-level recovery path can be added beneath it. The class states privilege only as given. Finding F-3.

6.7 §2.4 and §8.1 Consequence in Other Sectors

Held. The consequence arrived in the logical layer of answering points, hospitals, banks, airlines and agencies, and the physical facilities ran, as §2.4 states. Each consuming class owns its own reading; the class's note at §10.9 already carries the continuity point to them.

6.8 §6.3 The Governing Military-Action Scenario

Not exercised. The event was an accident. It supplies the consequence figures the scenario's activation half runs on (Section 10.2).

6.9 §5.4 Loss of the Signing or Distribution Infrastructure

Not exercised. The channel itself stayed available throughout.

6.10 §7.3 and §7.4 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 is recorded in the sources read as producing or gating the release.

6.11 Summary of Design Basis Performance

Of the ten readings, four Held, with a gap, the class's account among them; three Held; and three were Not exercised. Nothing was Missed. The class's §4.4 to §4.6, §5.1 and §7.2 were written from this event and read it. What the event adds is that a facility whose service cannot wait must not be in the first ring by default, that recovery is measured to the service and not the host, and that privilege is a parameter the design can lower.

7. Findings

Section references are to *Software, Hardware and Product Supply Chain* v0.1 unless another instrument is named.

7.1 F-1 — A Facility That Cannot Wait Is Not in the First Ring by Default

The content reached every customer in one session, and among the hosts it reached were those of answering points in Phoenix, Portland, Alaska and Middletown (Sections 3.3 and 3.5). The vendor's practice since lets customers choose to be among the first to receive content. The finding confirms §4.4's staging requirement and §6.8's second residual, and is that the class should state that the vendor's first ring is drawn so that a consuming facility whose service cannot wait (an answering point, a hospital, a control room) is not in it unless the facility chooses to be, and that the facility can always place itself in a later ring; and that the first ring's own loss is stated against the hosts it contains.

Affected: §4.4, §6.8. Disposition: R-1; noted to the consuming classes with the note at §10.9.

7.2 F-2 — Recovery Is Measured to the Service, Not the Host

The vendor reported about 99 percent of sensors online in ten days; one airline cancelled about 7,000 flights over five days while its hosts returned and its operation was rebuilt around them; hosts with encrypted disks needed recovery keys before they could be repaired (Sections 3.4 and 3.5). The finding confirms §4.6's requirement that the time to recover the most critical hosts be

stated and exercised, and is that the class should state the time to recovery of §3.3 in two parts, the hosts and the consuming service those hosts carry, and should require the consuming enterprise to state the second against its continuity window, counting the applications and schedules that must be rebuilt once the hosts return, and to hold its recovery keys where the same failure cannot reach them.

Affected: §3.3, §4.6, §10.1. Disposition: R-2.

7.3 F-3 — Privilege Is a Parameter the Design Can Lower

The sensor's kernel privilege is why a defect in its content stopped hosts at the level from which they boot, and why each had to be repaired by hand (Sections 3.1 and 3.4). On 19 November 2024 the operating-system vendor announced capabilities to let security products run outside the kernel and to apply fixes to machines that cannot boot (Section 3.6). The finding confirms §3.3's statement that kernel privilege sets whether a host can be recovered without hands, and is that the class should treat privilege as a design parameter to be held at the lowest level the product's function requires, with a product line stating why each function it runs at kernel or boot-path privilege needs it; and should credit a recovery path supplied by the platform beneath the product as satisfying §4.6 only where it needs neither the failed product nor the public networks.

Affected: §3.3, §4.6, §6.7. Disposition: R-3.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 to R-3 dispose to *Software, Hardware and Product Supply Chain*.

No.	Title	Affected	Recommendation	Pri.
R-1	Keep facilities that cannot wait out of the first ring	§4.4, §6.8	Draw the first ring so that answering points, hospitals and control rooms are not in it unless they choose; let any facility place itself later; state the first ring's loss. Noted to the consuming classes.	1
R-2	Measure recovery to the service	§3.3, §4.6, §10.1	State time to recovery for hosts and for the service they carry, against the continuity window; hold recovery keys beyond the same failure.	1
R-3	Lower privilege by design	§3.3, §4.6, §6.7	Hold privilege at the lowest level the function needs and state why each kernel function needs it; credit a platform recovery path only if independent of the product and the public networks.	2

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2 and the impacted sectors. OI-2: the class has not taken in this instrument and conforms at its next revision, with Erratum 2. OI-3: the airline's litigation is open; nothing here rests on its outcome.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding of fault.

10.2 Relationship to the Class's Use Case

The class's use case traces §6.3, in which a pre-positioned capability is activated as a destructive update during an attack on the grid. This event supplies the figures for the activation half as an accident produced them: a release window of seventy-eight minutes; about 8.5 million hosts stopped from the kernel; hosts that the channel could not correct; recovery by hand with recovery keys, to about 99 percent of sensors in ten days; and a consuming operation that took five days to rebuild. F-2 and F-3 bear on how long the use case's facilities wait, and on what the use case may credit for their recovery when, unlike here, the hands are needed elsewhere and the networks may be down.

10.3 Relationship to Other Instruments

The Compromise of the SolarWinds Orion Platform (DBA-IT-RE-1) and *The Backdoor in the XZ Utils Compression Library* (DBA-IT-RE-3) read the integrity chain through the same kind of channel. The answering points' manual working bears on *Public-Safety Answering and Dispatch* and is read for a ransomware attack in *The Ransomware Attack on Baltimore's Computer-Aided Dispatch* (DBA-ESS-RE-4).

References

CrowdStrike, *Preliminary Post Incident Review*, July 2024; *Channel File 291 Incident Root Cause Analysis*, 6 August 2024; remediation and guidance hub. Microsoft, statement of 20 July 2024 on the number of devices affected; *Windows security and resiliency: Protecting your business*, 19 November 2024. House Committee on Homeland Security, Subcommittee on Cybersecurity and Infrastructure Protection, *An Outage Strikes: Assessing the Global Impact of CrowdStrike's Faulty Software Update*, 24 September 2024, Serial No. 118-81. Congressional Research Service, *CrowdStrike IT Outage: Impacts to Public Safety Systems and Considerations for Congress*, IF12717, version of 4 December 2024. Reports of the Department of Transportation's closing of its review of Delta Air Lines, 2026, and of the order of the Superior

Court of Fulton County, Georgia, of May 2025. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 14 October 2026: issued in Project 11 under the Information Technology Sector Framework v0.5, on the conferral of the framework's §9.1 of 14 October 2026, read against *Software, Hardware and Product Supply Chain* v0.1. Written new under the Reference Event Compression Standard v0.3. Tier determination at Section 5.2; F-1 to F-3, R-1 to R-3. The class's account holds, with two qualifications going to the class as Erratum 2 of Project 11's note of 14 October 2026 (Information Technology batch).